

Executive Technical Recommendation: Optimal pH for Adsorption of 2-Naphthoxyacetic Acid

The adsorption performance of the treatment system was tested at pH 3 and pH 9 to find the best operating condition for removing 2-naphthoxyacetic acid from aqueous solutions. The compound is a weak acid with a pKa of 3.55, so the solution pH controls whether it sits in its protonated, neutral form or its deprotonated, negatively charged form. Using the Henderson-Hasselbalch relationship, at pH 3 (below the pKa) the molecule is mostly protonated and neutral, about 78% protonated and 22% deprotonated. At pH 9 (well above the pKa) it is almost fully deprotonated and anionic, close to 100%. This matters because neutral molecules tend to form stronger hydrophobic and pi-pi interactions with the adsorbent surface. The anionic form is more strongly solvated in water and can be pushed away electrically, so it adsorbs less.

The analytical wavelength of 325 nm was used for quantitative tracking, even though it is not the tallest peak in the raw scan. The full absorbance profile in Kinetic data.xlsx shows an even bigger signal near 271 nm, running about 2.0 to 2.9 absorbance units across the dataset at both pH conditions. That is well outside the range shown in the UV-Vis spectra images. Readings that high fall outside the instrument's reliable linear range for Beer-Lambert work, where stray light and detector nonlinearity distort the link between absorbance and concentration, so a tall peak there is not something we can trust for quantitation. The 325 nm peak, in contrast, is well resolved and stays inside a linear working range (absorbance about 0.7 to 1.15 across the dataset) at both pH conditions, which makes it the dependable choice. It was used for both conditions because its position does not move with pH.

Adsorption kinetics showed fast initial uptake that slowed down over time, but neither condition had leveled off by the final reading at 460 minutes. The measured adsorption capacity reached 57.54 Mol/g at pH 3 versus 39.96 Mol/g at pH 9, and the adsorption efficiency reached 35% at pH 3 versus 23% at pH 9. So at the last reading, pH 3 beats pH 9 by about 44% in capacity and about 51% in efficiency. Because neither condition had leveled off, the pseudo-second-order uptake model was fitted to each capacity-time series to estimate the true saturation point. At pH 3, the fitted equilibrium capacity is 103.5 Mol/g, with a rate constant of 2.58×10^{-5} g/(Mol·min) and R^2 of 0.995. At pH 9, the fitted equilibrium capacity is 106.1 Mol/g, with a rate constant of 1.14×10^{-5} g/(Mol·min) and R^2 of 0.991. So the fitted equilibrium capacities are actually close to each other, within about 2.5%, unlike the roughly 44% gap at the last reading. The real difference between the two conditions is kinetic, not a capacity ceiling. pH 3's rate constant is more than double pH 9's, so pH 3 reaches a much bigger share of its ultimate capacity within a practical contact time: 56% of its fitted equilibrium by 460 minutes, versus 38% at pH 9.

The treatment system should be operated at pH 3. The two conditions may reach a similar ultimate capacity given enough contact time, but pH 3 is held largely in its neutral, protonated form (about 78%), and that form drives the stronger surface interactions behind its much faster uptake and its higher measured capacity and efficiency at any realistic operating timescale. Operating at pH 9 pushes the molecule almost entirely into its repelled anionic form (close to 100%), which measurably slows adsorption and lowers performance within the timeframes this system will actually run.